

Parallel Robots

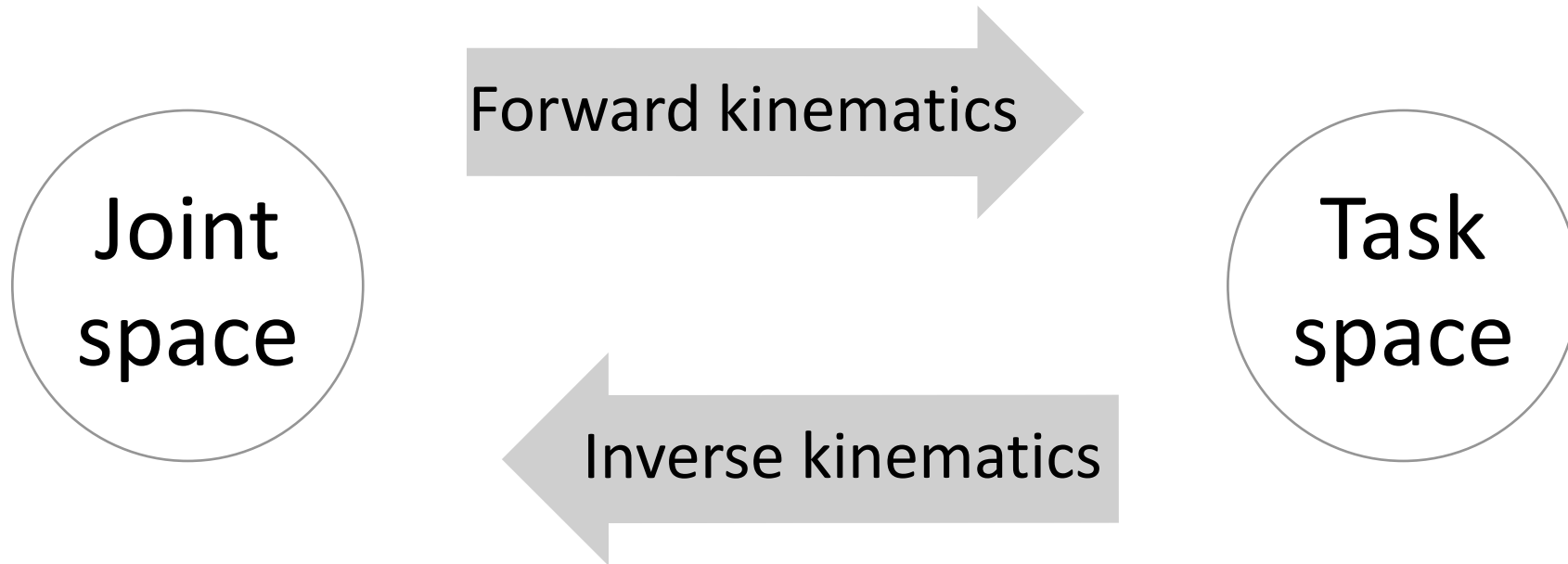
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Forward and Inverse Kinematics

- Goal: establishing the relations between joint-space coordinates and task-space coordinates



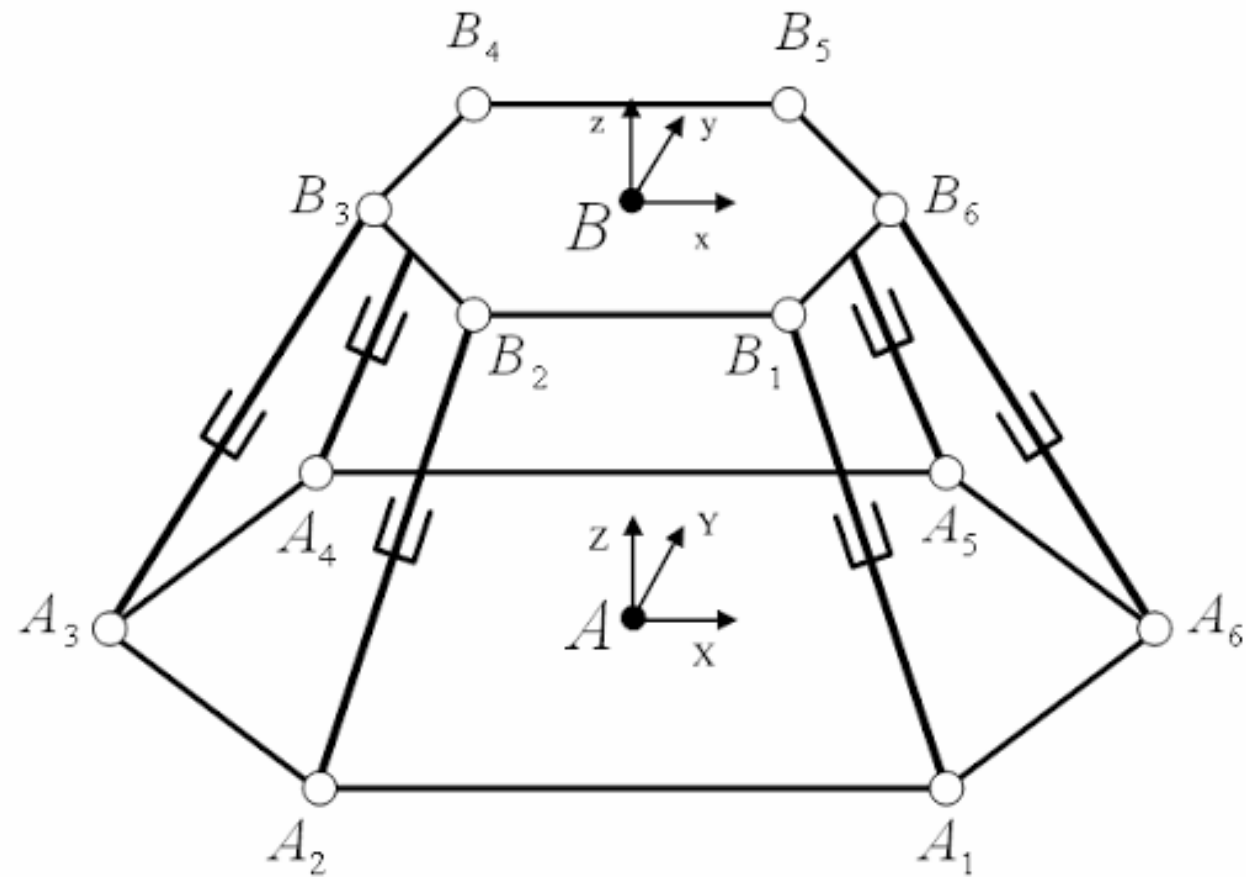
- 2-RPR robot
 - How to actuate?
 - How to parametrize? $\rightarrow$ joint-space and task-space coordinates
 - How many assembly modes?
 - How many working modes?

- Joint-space coordinates: $\mathbf{q} \in \mathbb{R}^n$ or $\mathbf{q}_a \in \mathbb{R}^n$
 - Actuated joint coordinates, describing an angle or a signed length
- Passive joint coordinates: $\mathbf{q}_p \in \mathbb{R}^{n_p}$
- Task-space coordinates: depends on the robot and on the task
 - Generalized coordinates: $\mathbf{x} \in \mathbb{R}^m$ with $m \geq n$ (why?)
 - Position, Euler angles, quaternions, rotation matrices...
 - Homogeneous transformation: $\mathbf{T} \in SE(3)$
 - How many independent coordinates? How many constraint equations?
 - What about redundant robots?

- Main goal of geometric modeling
 - Defining the robot configuration from a given set of coordinates
- Common geometric models
 - Implicit: $f(\mathbf{q}, \mathbf{x}) = 0$
 - Inverse (IK): $\mathbf{q} = f_{IK}(\mathbf{x})$
 - Forward (FK): $\mathbf{x} = f_{FK}(\mathbf{q})$
- Main approach: loop-closure equations

- Goal: finding joint-space coordinates from task-space coordinates: $\mathbf{q} = \mathbf{f}_{IK}(\mathbf{x})$
- Why?
- On parallel robots, IK are usually simple $\rightarrow$ closed-form solution
 - Why?
- Working modes = solutions of the IK
- How to choose the working mode?

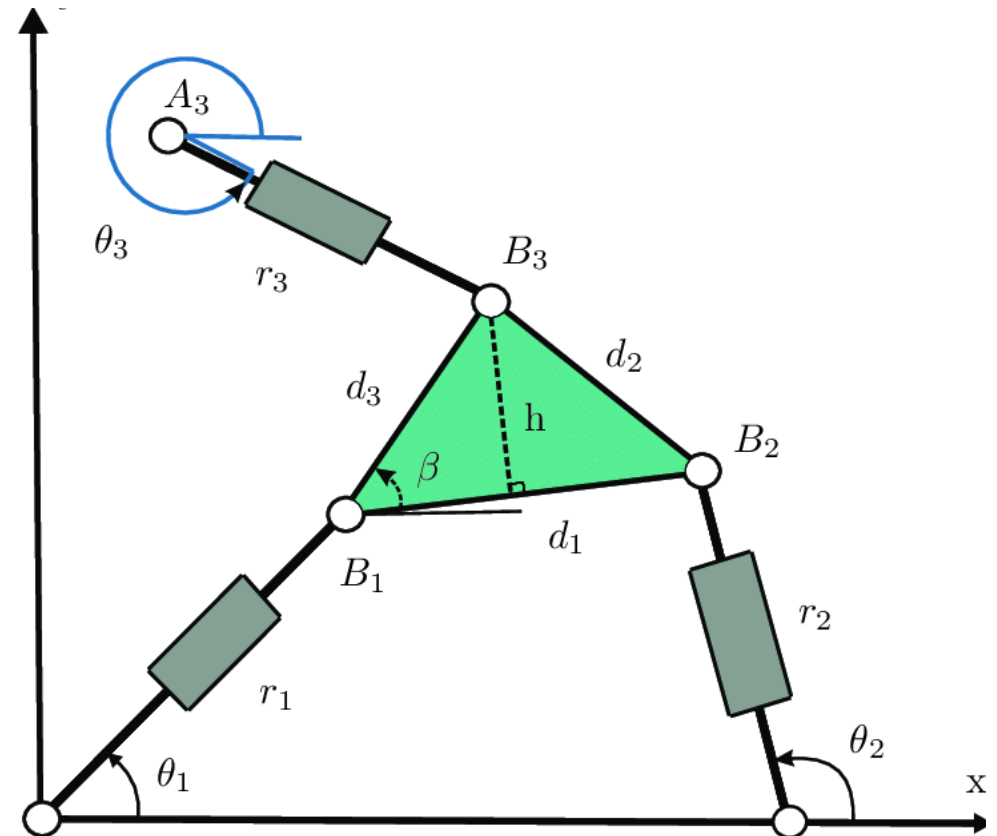
- Example: Gough-Stewart platform



<https://doi.org/10.1115/IMECE2009-11148>

- Goal: finding task-space coordinates from joint-space coordinates: $\mathbf{x} = \mathbf{f}_{FK}(\mathbf{q})$
- Why?
- On parallel robots, FK usually have no closed-form solution
 - What to do?
- Assembly modes = solutions of the FK
- How to choose the assembly mode?

- Example: 5-bar mechanism
 - Analytical solutions
- Example: 3-RPR robot
 - Analytical solutions → exercise
- Example: Gough-Stewart platform
 - No known analytical solution
 - Up to 40 distinct solutions



<https://doi.org/10.1016/j.mechmachtheory.2018.05.004>

- There are several solutions to FK
- How to choose the assembly mode corresponding to the actual platform pose?
 - Sensor redundancy
 - Exteroception
 - Numerical methods with theoretical guarantees
 - E.g., interval methods, Kantorovitch's theorem

- Newton's methods (or Newton-Raphson method)